

Distributed Network Control of Multi-UAV Systems for Cooperative Heavy-Load Transport Using a Virtual-Passivity Framework

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Abstract—This paper presents a novel distributed network control framework for cooperative heavy-load transportation using multi-UAV systems, accounting for thrust limitations and heterogeneous cable characteristics. By constructing a virtual passive system comprising interconnected virtual nodes, springs, and dampers, the proposed method decouples internal coordination stability from external velocity tracking. A velocity tracking controller is devised to asymptotically steer the load's velocity toward a desired trajectory, while preserving inter-agent cohesion through virtual interactions. Notably, the controller operates without explicit inter-UAV communication, relying solely on relative position measurements. Numerical simulations involving ten UAVs transporting a 14 kg load—exceeding 76% of their combined thrust capacity—along a figure-eight trajectory validate the proposed method. Field tests with six UAVs transporting a 6 kg load are conducted to validate the control framework's performance in practical applications. The results confirm accurate velocity tracking, balanced cable tension distribution, and scalability to heterogeneous UAV team configurations.

Index Terms—Cooperative transport, distributed network control, low-altitude economy, passivity-based control, UAV swarms.

I. INTRODUCTION

THE low-altitude economy, characterized by operations of manned and unmanned aerial vehicles (UAVs) within airspace below 1000 meters, is rapidly emerging as a transformative paradigm for applications such as smart logistics, infrastructure inspection, and emergency response in both urban and rural settings [1], [2], [3], [4], [5], [6], [7]. Regarded as a major driver of future economic growth, this sector holds substantial and expansive development prospects [8], [9], [10]. However, the successful commercialization of these services hinges on the development of robust, scalable, and safe aerial systems capable of reliable operation within complex and dynamic low-altitude environments [11], [12], [13], [14], [15].

The evolution of the low-altitude economy faces a critical bottleneck in that the weight of payloads can vary significantly depending on specific operational requirements, particularly

when dealing with heavy payloads such as construction materials or large industrial components [16], [17], [18]. This variability often exceeds the capacity of a single UAV, rendering conventional UAV designs insufficient for the demands of heavy-load logistics which is a cornerstone of the low-altitude economy's value proposition. While suitable for lightweight tasks, single UAVs prove inadequate for these more demanding applications, necessitating innovative solutions to address this fundamental limitation [19]. Two primary approaches have emerged to tackle this challenge: the development of UAVs with higher individual payload capacities, and the implementation of multiple UAVs working cooperatively to share loads [20], [21]. The first approach, while technically viable, requires substantial investment in terms of time and resources for developing new UAV models and carries significant economic implications as the cost of heavy-lift UAVs tends to increase in a geometric progression relative to their payload capacity [17].

The second approach, employing multi-UAV cooperative transport systems, presents a particularly compelling solution that offers enhanced lift capacity through collective effort, improved reliability through built-in redundancy, and greater mission flexibility through adaptive coordination [22], [23]. This approach has garnered significant attention for heavy-load logistics in unstructured settings such as power grid construction, disaster relief operations, and large-scale agricultural spraying, where traditional ground-based systems face limitations in accessibility and operational efficiency [3], [5], [16]. The distributed nature of multi-UAV systems provides inherent advantages including enhanced operational flexibility, increased deployment speed, and improved fault tolerance through redundancy which are all critical factors for scaling operations in dynamic low-altitude scenarios [24].

Multi-UAV cooperative transport can be categorized based on the connection mechanism into rigid connections and soft connections [25], [26]. Rigid connections typically involve mechanical components such as robotic arms, clamps, and magnets [27], [28]. The primary advantage of rigid connections lies in their ability to actively grasp payloads through devices like clamps or robotic arms, which effectively suppress oscillations and provide higher controllability with increased degrees of freedom [29]. However, these systems have some drawbacks, including high design complexity, increased cost, greater weight and energy consumption, and elevated rotational inertia, which can negatively impact UAV maneuverability [30]. In contrast,

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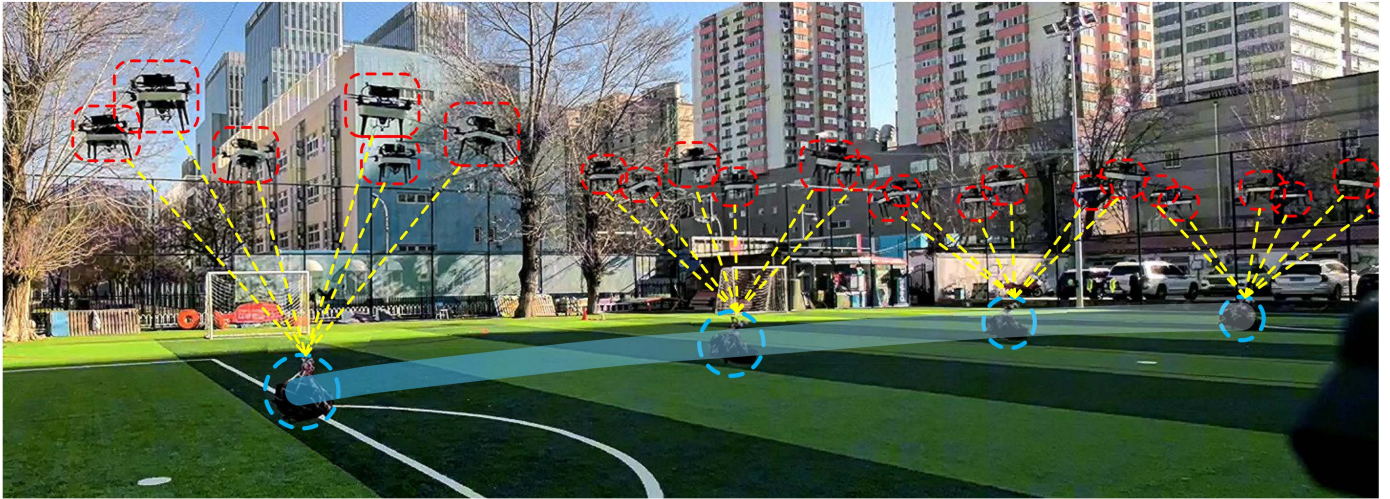


Fig. 1. Outdoor experiment demonstrating cooperative transport with six UAVs. The red dashed boxes represent the UAVs, the blue dashed box denotes the payload, and the bold yellow dashed lines indicate the cables connecting the UAVs to the cargo. The thick blue gradient line illustrates the trajectory of the payload. The payload has a mass of 6 kg, while each UAV has a sustained load capacity of approximately 1.6 kg.

soft connections, typically realized through cables, offer several advantages, including lower cost, simplicity, lightweight construction, reliability, and ease of rapid deployment [23], [31], [32]. Additionally, soft connections do not require precise matching between the load shape and the UAV configuration, making them more versatile for diverse applications [33], [34].

However, the fundamental challenge of soft connections, such as cables, lies in their inherent unilateral actuation constraint: they can only exert tensile forces along their axial direction, but cannot resist compressive loads or bending moments in other directions [19], [32], [35]. This limitation leads to a highly underactuated system that is prone to oscillations and presents significant control difficulties [34]. Moreover, when coordinating multiple UAVs for collaborative transport of a single suspended payload, substantial complexities emerge, which are exacerbated in dense and unpredictable low-altitude airspace, such as urban or industrial environments. These complexities include avoiding dynamic obstacles, compensating for sudden wind disturbances, and mitigating communication delays or dropouts that can disrupt real-time cooperative control [13], [14], [36]. Consequently, the successful deployment of such systems demands advanced control strategies and robust architectural frameworks to ensure both operational safety and efficiency under challenging conditions, making this a key area of research in aerial robotics and distributed systems [37], [38]. This paper primarily focuses on the challenges and solutions related to multi-UAV cooperative transportation using soft connections, specifically cable-based systems.

The scalability and reliability of multi-UAV systems for collaborative transport of a single suspended payload hinge on distributed network control strategies that can address fundamental challenges in network science and engineering [39], [40], [41]. These include communication constraints (e.g., bandwidth limitations and packet loss in GPS-denied environments), heterogeneous agent dynamics (e.g., varying UAV capabilities and cable lengths), and environmental uncertainties (e.g., wind gusts and payload variations) [42], [43], [44], [45], [46], [47].

Addressing these issues effectively requires an interdisciplinary integration of network science and UAV control theory, which will in turn accelerate the deployment of such systems in the low-altitude economy and support its sustainable development [48]. To address these communication challenges, this paper proposes a distributed control framework that relies solely on relative position measurements, thereby eliminating the dependency on explicit inter-UAV communication and enhancing robustness in communication-limited environments.

Current multi-UAV cooperative transport strategies typically fall into three categories: formation-based control, planning-based control, and admittance control [25], [26]. Formation-based approaches are computationally efficient but lack mechanisms for cable tension regulation and are incapable of accommodating uncertainties in cable lengths [42], [49], [50]. Planning-based methods enable precise tension distribution and high-fidelity trajectory tracking but are hindered by model complexity, scalability issues in large formations, and susceptibility to single-point failures [33], [51], [52], [53]. Admittance control leverages active motion from leader UAVs with follower adjustments based on estimated tension; however, it frequently leads to load imbalances, such as leader overload and follower cable slack, introducing hazardous slack-to-taut transitions that threaten system stability [32], [54]. Despite significant advancements in multi-UAV cooperative transport research, most existing studies remain confined to indoor environments equipped with motion capture systems [34], [49], [51], [52], [54], thereby overlooking the impact of complex outdoor disturbances and environmental uncertainties. Moreover, the experimental loads are often lighter than the capacity of a single UAV [49], [51], [54], which undermines the fundamental aim of extending cooperative capabilities to heavy-load scenarios. Therefore, the development of computationally efficient and high-load transport methods that are robust against thrust limitations and cable-length variability remains a pressing challenge.

To address this gap, we propose a virtual-passivity-based distributed network control framework for multi-UAV

heavy-load transport, extending our previous work [55]. In our prior work [55], we introduced a dissipation-based cooperative transport system that is composed of virtual nodes, a virtual tray, virtual springs, and dampers. This transport system is robust to uncertainties in cable length and load capacity, and it supports plug-and-play. However, it was limited to zero-input stability analysis and empirical validation for point-to-point transport, lacking theoretical guarantees. For applications requiring obstacle avoidance or high-precision trajectory following, achieving accurate cooperative velocity tracking remains essential. This research has improved the dissipation-based cooperative transport system to a virtual-passivity-based system, which decouples internal network stability from external velocity tracking. The virtual passive system has been proved rigorously to exhibit strict output passivity, ensuring finite-gain $\mathcal{L}_2$ stability. The proposed cooperative transport framework guarantees asymptotic convergence for a bounded first-order differentiable desired velocity. This design aligns with network control principles by leveraging passive systems theory to guarantee scalability and disturbance rejection without inter-UAV communication [56], [57], [58], making it ideal for low-altitude networks with limited bandwidth and dynamic environments. The primary contributions of this work are multifaceted:

- 1) *Distributed Network Control Framework*: We develop a novel control architecture that decouples internal stability maintenance from external trajectory tracking. This framework operates solely on relative position measurements, eliminating the need for explicit inter-UAV communication and reducing computational overhead to linear complexity with swarm size. It explicitly addresses thrust limitations, cable-length uncertainties, and external disturbances, which are critical for heavy-load scenarios.
- 2) *Rigorous Theoretical Analysis*: We provide comprehensive proofs showing that the virtual passive system is strictly output passive, ensuring zero-input asymptotic stability and finite-gain $\mathcal{L}_2$ stability. Additionally, the proposed velocity tracking controller guarantees asymptotic convergence for bounded, first-order differentiable velocity profiles, with theoretical bounds on tracking errors (e.g., errors under 0.1m/s in simulations).
- 3) *Experimental Validation*: We validate the framework through extensive numerical simulations (involving ten UAVs transporting a 14 kg load, exceeding 76% of their combined thrust capacity) and outdoor field tests (with six UAVs transporting a 6 kg load). Results demonstrate balanced cable tension distribution, accurate velocity tracking, and scalability to heterogeneous UAV teams, highlighting practical applicability in low-altitude economy settings.

This study aligns with the vision of autonomous, dependable, affordable network systems for the low-altitude economy [59], [60]. By integrating virtual passivity with distributed control, we advance the state-of-the-art in network science for UAV swarms, offering a scalable solution for future smart logistics. The remainder of this paper is organized as follows: Section II reviews related works, Section III outlines the problem formulation, Section IV details the controller design, Section V

presents simulations, Section VI analyses experimental results, and Section VII concludes the paper.

II. RELATED WORK

The rapid advancement of low-altitude economy operations has catalyzed significant research efforts in aerial transportation systems, particularly focusing on the development of robust and scalable multi-UAV cooperative transport frameworks [10], [62], [63]. This section provides a comprehensive review of the state-of-the-art in single and multiple aerial vehicle transportation systems, examining fundamental modeling approaches, control strategies, and network integration methods that form the foundation for cooperative heavy-load transportation. The review systematically analyzes three key areas: single aerial vehicle transport research, multi aerial vehicle cooperative transport systems and the integration of network science principles. By analyzing existing methodologies and identifying persistent research gaps, this paper establishes the theoretical and practical context for the virtual-passivity-based distributed control framework, particularly highlighting the need for solutions that address thrust limitations, cable-length uncertainties, and communication constraints in real-world heavy-load transport scenarios.

A. Single Aerial Vehicle With Suspended Load Transport Research

Single aerial vehicle transport serves as the foundation for multi-UAV systems, with studies covering modeling and control strategies. Research in this area has evolved from basic dynamics modeling to advanced control techniques, addressing issues such as payload swing and environmental disturbances [19], [64], [65].

1) *Modeling Approaches for Single-UAV Slung-Load Systems*: Early research on single aerial vehicle transport with suspended load transport focused on dynamics modeling using Euler-Lagrange or Newton-Euler methods. For instance, Lucassen et al. [66] derived motion equations for helicopters with slung loads, revealing swing stability issues and the impact of cable length on system dynamics. Fusato et al. [67] extended these models by incorporating rotor dynamics and payload interactions, achieving higher fidelity at the cost of increased computational complexity. Meanwhile, research pivoted toward enhancing real-time state estimation through sensor fusion. Lee et al. [68], for instance, demonstrated accurate swing angle estimation using onboard IMUs and tension sensors, eliminating the dependency on external motion capture systems. Kaan Sansal et al. [69] developed a high-fidelity model of a light utility helicopter with a slung load to analyze their dynamic interactions. By performing stability analysis on linearized models across different flight conditions, they quantified how key parameters like cargo mass, cable length, and attachment points affect the damping and stability of the coupled system. Although the modeling literature for single-aircraft slung-load systems serves as a critical precursor, the transition to cooperative multi-UAV transport introduces significant complexity due to its multi-body

and over-constrained nature, requiring the implementation of other analytical tools.

2) *Control Strategies for Single-UAV Slung-Load Systems:* Control strategies for single aerial vehicle transport with suspended loads range from traditional PID to advanced methods like adaptive control and model predictive control (MPC). Omar [70] designed a fuzzy-based anti-swing controller that added compensatory displacements to UAV trajectories, effectively reducing oscillations in simulation environments. Sreenath et al. [71] utilized differential flatness theory for trajectory tracking, enabling precise control of both UAV and load states. Yang et al. [72] proposed an energy-based nonlinear controller that compensated for unknown cable lengths, demonstrating robustness in outdoor tests. A notable systematic solution further combines real-time planning under dynamic feasibility constraints with an adaptive NMPC enhanced by a hierarchical disturbance compensator, effectively addressing unknown perturbations and model inaccuracies during aggressive flight in constrained environments [73]. However, many of these methods rely on external sensing systems, such as VICON motion capture, limiting their applicability in GPS-denied or outdoor environments. Lv et al. [74] incorporated disturbance observers (DOBs) to estimate external forces, enhancing adaptability to wind and payload variations. Despite these progresses, single-UAV systems remain constrained by thrust limits, prompting a shift to multi-UAV approaches for heavy-load applications.

B. Multi-UAV Cooperative With Suspended Load Transport Research

Multi-UAV systems enhance payload capacity and robustness through distributed networks, but introduce complexities in coordination, communication, and control. Research in this area spans modeling, formation control, trajectory planning, and admittance control, each addressing specific aspects of cooperative transport.

1) *Modeling Approaches for Multi-UAV Cooperative With Suspended Load Systems:* The fundamental challenge in modeling cooperative UAVs-load systems lies in accurately capturing the cable dynamics that govern load behavior and overall stability [75]. Early research favored simplified representations, particularly the massless rigid-link model, which treats cables as inextensible bars maintaining fixed UAV-payload distances. This approximation reduces the system to pendulum-like dynamics described by point masses and swing angles, offering high computational efficiency. Michael et al. [76] applied this model to 3D manipulation using inverse kinematics, while Sreenath et al. [77] validated its practicality in trajectory planning under the taut-cable assumption. However, such models fall short in aggressive maneuvers where cable slackness and elasticity dominate, motivating the development of more refined frameworks.

Elastic cable models improve fidelity by incorporating spring-damper elements to capture oscillatory behavior. Kotaru et al. [78] introduced damping and stiffness parameters to better represent oscillations during dynamic flight, and Pizetta et al. [79] extended this to dual-UAV systems with traction-based disturbance rejection. Though more physically accurate, these

models incur higher computational cost, complicating real-time implementation.

For higher precision, distributed-mass models discretize the cable into multi-segment chains. Goodarzi et al. [80] used weighted links connected by spherical joints to analyze cable elongation and vibration. Rastgoftar et al. [81] further advanced this concept with “tensegrity muscles” applying tension and compression to form reconfigurable structures in constrained spaces. Such models capture nonlinear effects effectively but remain computationally prohibitive for online control.

Despite these advances, key challenges remain in multi-UAV cable-load modeling. Transient regimes, during takeoff, landing, or cable collisions, are often oversimplified [82]. High-fidelity models remain computationally expensive, leading to approximations like height-based slack-taut switching [83]. Experimental validation under realistic outdoor conditions with wind, uncertainty, and external disturbances is still scarce [84], underscoring the need for robust, experimentally-grounded modeling frameworks.

2) *Formation-Based Control for Multi-UAV Cooperative With Suspended Load Systems:* Formation-based control addresses cooperative transport as a consensus problem, in which UAVs maintain predefined relative positions to achieve desired load motion. Early work by Bernard et al. [4] employed sensor-based coordination for triple-UAV systems with robust control to mitigate wind disturbances, though the method relied on continuous communication and exhibited limited performance under highly uncertain conditions. Zhang et al. [42] later introduced a self-triggered cooperative control framework that reduces communication frequency by discretizing update events, enabling effective path following in outdoor experiments. Nevertheless, such communication-dependent strategies face challenges in dynamic low-altitude settings with obstacles and GPS denial, where reliable relative positioning is essential. Recent advances, such as the dissipativity-based method by Liu et al. [50], have improved robustness by enabling formation stability using only relative measurements, without explicit communication.

Despite their advantages in computational efficiency and operational simplicity, formation-based methods present several noteworthy limitations. They often treat cable tension as an external disturbance rather than an actively optimized variable, which limits the ability to distribute load capacity optimally among UAVs. Additionally, their emphasis on maintaining rigid geometric formations compromises adaptability in cluttered or confined environments. More critically, these approaches generally lack inherent robustness to system anomalies such as individual UAV failures, motor saturations, or uncertainties in cable length [42], [49], [50], [51], [85]. These shortcomings motivate the development of more flexible and fault-tolerant control architectures.

3) *Trajectory Planning-Based Strategies for Multi-UAV Cooperative With Suspended Load Systems:* Trajectory planning-based methods for multi-UAV cooperative transport generate individual trajectories for each UAV by modeling payload, cable, and vehicle dynamics, followed by path-tracking control. These approaches generally rely on inter-agent communication to resolve trajectory conflicts and maintain coordination.

Kumar's team [86] leveraged differential flatness to plan feasible trajectories for load position and orientation control, validating the method experimentally with three quadrotors. Subsequent work [32] extended this to cooperative transport using full dynamic models, without quasi-static assumptions, for both point-mass and rigid-body payloads. The resulting systems were shown to be differentially flat under positive cable tension, while scenarios allowing non-negative tension (including slack cables) were modeled as differentially flat hybrid systems due to tension-induced switching dynamics. Gassner et al. [31] combined PD and LQR control for aggressive maneuvers without explicit communication, though dependence on onboard vision limited outdoor use. Domestic researchers such as Wang [87] introduced distributed trajectory optimization for cluttered settings, using parallel computation to scale to large teams. Nevertheless, such methods often incur high computational costs, with planning complexity growing significantly with the number of UAVs. Jackson et al. [88] tackled scalability via distributed optimization, enabling real-time trajectory generation for cable-suspended loads, though the approach remains computationally demanding for resource-constrained platforms.

When accurate system modeling is achievable, trajectory-based methods enable precise trajectory tracking and effective cable tension distribution [21], [33], [51], [52], [53]. However, they face three major limitations: First, model complexity leads to computational demands that scale sharply, often exponentially, with the number of UAVs. Second, trajectory optimization becomes ill-conditioned under numerous constraints (e.g., collision avoidance and physical connectivity), causing numerical instability. Third, these approaches exhibit limited inherent robustness to system uncertainties such as UAV failures or cable-length variations.

4) *Admittance-Based Control for Multi-UAV Cooperative With Suspended Load Systems*: Admittance-based cooperative transport in multi-robot systems typically operates under a leader-follower framework, in which follower robots employ admittance control to dynamically adjust their motion in response to estimated external forces. Followers rely solely on onboard IMUs and system models to estimate these forces, continuously modifying their reference trajectories via an admittance controller to maintain coordination without explicit inter-robot communication.

Tagliabue et al. [89] introduced a passive force control strategy for two hexacopters transporting large objects (Fig. 16). Inspired by biological cooperation, their leader-follower approach used an Unscented Kalman Filter (UKF) to estimate external forces on the follower drone via a visual-inertial navigation system, eliminating the need for communication, payload shape knowledge, or grasp point locations. However, the experiments were conducted indoors without considering outdoor conditions such as weather or varying lighting, indicating limited practical applicability. Similarly, the team from Hong Kong City University [90] developed a leader-follower framework integrated with Kalman filtering to reduce communication dependency, achieving stable formation in indoor settings. In contrast, Romano [35] designed a haptically-guided cooperative system where five quadrotors tethered to a payload responded to human-applied

forces estimated via inline tension sensors. These forces updated the virtual dynamics of an admittance controller, generating trajectory commands that were distributed by a central payload controller and executed by open-source autopilots. This system was successfully validated in outdoor flight tests.

Admittance control faces notable limitations in real-world deployments. When a follower estimates near-zero external force, indicating slack, its admittance controller becomes ineffective, leaving the UAV in hover until cable tension is restored. Meanwhile, leader robots operating at full thrust capacity may fail in trajectory tracking, approaching instability. These issues highlight the need for more adaptive and fault-tolerant control strategies to ensure safety and performance under uncertain conditions [32], [35], [54].

C. Network Science in Multi-UAV Cooperative Transport Research

The integration of network science principles into multi-UAV cooperative transport has facilitated the systematic design and analysis of communication, coordination, and control strategies for UAV swarms [91]. Graph theory, in particular, offers a powerful framework for modeling interaction topologies and evaluating system-level properties—such as connectivity, robustness, and efficiency—in these large-scale cyber-physical systems [92].

A central theme in this domain is the design of network topologies. Early work by [93] explored how graph structures influence formation stability and control performance, laying a foundation for topology optimization based on mission requirements. Subsequent studies on cooperative transport have extended these ideas, showing that formation graphs, inter-vehicle distances, and load-attachment geometry collectively govern system stability and tracking accuracy [49], [94]. Further research on cable-suspended and slender-payload systems confirms that well-designed communication and coordination topologies can substantially improve robustness and maneuverability in complex manipulation tasks [95].

Consensus algorithms have played a key role in enabling distributed coordination without centralized control. Recent protocol-based schemes explicitly accommodate limited bandwidth, asynchrony, and packet loss, allowing UAV teams to reach agreement on shared variables—such as formation shape or transport velocity—using only local communication [96]. These methods are supported by data-driven stability analyses that provide theoretical guarantees for consensus under real-world conditions [97]. Consensus mechanisms have also been incorporated into distributed task allocation algorithms, improving scalability in heterogeneous networks [98], while reinforcement learning has been used to adapt consensus and formation policies to uncertain network dynamics [99].

Robustness to network disruptions remains a critical concern. In cooperative transport, adaptive controllers have been developed to handle modeling errors, wind disturbances, and unmodeled payload dynamics while maintaining swing attenuation and formation cohesion [34], [100]. Human-robot interaction

TABLE I
RELATED WORKS OF MULTI-UAV COOPERATIVE TRANSPORT IN LOW-ALTITUDE ECONOMY

Reference	Approach	Contributions	Limitations
[42]	Formation-based control	(1) Proposes a distributed cooperative control scheme that significantly reduces communication requirements (2) Conducts experimental verification with three UAVs in an outdoor environment	(1) Requires GPS availability and communication links between UAVs (2) The payload mass is lighter than the capacity of a single UAV (3) Does not address cable length uncertainty, balanced tension distribution, or fault tolerance capability
[49]	Formation-based control	(1) Develops a robust adaptive formation control methodology for UAVs subject to bounded exogenous disturbances (2) Employs a σ -modification approach instead of Model Reference Adaptive Control (MRAC) techniques to handle unknown disturbances	(1) Relies on networking and high-frequency communication between UAVs and ground stations (2) Experiments are conducted only in indoor controlled environments using precise LPS-based external positioning systems (3) Does not consider balanced tension distribution or fault tolerance mechanisms
[61]	Admittance-based control	(1) Presents a method for transporting bulky objects without explicit communication between agents (2) Employs robust control theory to quantify parametric uncertainties and unmodeled dynamics, ensuring stability through proper parameter selection	(1) Indoor experiments do not account for real-world factors such as wind and lighting conditions that may affect visual sensors and IMU performance (2) Does not address balanced tension distribution or fault tolerance capability (3) Exhibits limited scalability to larger-scale UAV teams
[54]	Admittance-based control	(1) Implements a leader-follower framework that operates without inter-UAV communication (2) Develops Unscented Kalman filters (UKFs) to estimate the forces applied by both the leader and the follower UAVs	(1) Experiments are conducted indoors and rely on OptiTrack motion capture systems (2) Shows limited scalability to larger UAV teams (3) Does not incorporate fault tolerance mechanisms
[52]	Trajectory Planning-based	(1) Utilizes an A*-APF hybrid algorithm to achieve safe and effective cooperative suspended transport (2) Enables precise trajectory tracking and tension distribution for dual-UAV systems	(1) Computational complexity limits scalability to larger UAV teams (2) Experiments are conducted indoors with light payloads (3) Does not address cable length uncertainty or fault tolerance capability
[21]	Trajectory Planning-based	(1) Proposes a framework that solves the whole-body kinodynamic motion planning problem online while accounting for dynamic coupling effects and constraints (2) Demonstrates high maneuverability with agile obstacle avoidance, narrow gap traversal capabilities, and robust wind disturbance resistance	(1) The whole-system modeling and solving approach leads to a computational explosion when scaling to larger UAV teams (2) The trajectory computation employs a centralized architecture, relying on high-performance ground stations and networked communication (3) Underestimating the load mass and inertia can lead to violations of the maximum thrust constraints, and the method does not account for fault tolerance
This work	Virtual-passivity distributed network control	(1) Operates using only relative position measurements, without inter-UAV communication and reduces computational overhead to linear complexity with swarm size (2) Addresses thrust limitations, cable-length uncertainties, external disturbances, and fault tolerance, which are critical for heavy-load scenarios	(1) Lacks environmental perception capabilities and obstacle avoidance competence (2) Relies on external sensors for agent localization (e.g., GPS in our experiments), but the control law requires only relative position measurements, enabling potential integration with GPS-free relative sensing techniques.

scenarios further underscore the need for resilience to both communication faults and physical disturbances [101]. From a network perspective, recent studies on UAV dependability analyze performance degradation under node or link failures and summarize mitigation strategies—such as redundancy, re-routing, and fault-tolerant consensus—for safety-critical operations [102].

To manage complexity in large-scale deployments, multi-layer network frameworks have been proposed. These hierarchical architectures decouple sensing, communication, and control into separate layers, enabling modular system design. In such frameworks, the communication layer manages aerial link routing and resource allocation, while the control layer executes formation-keeping and transport algorithms [103]. This abstraction supports scalability and integration with ground networks and edge computing infrastructure [92].

Security and resilience have also been informed by network science. Dependability studies identify threats such as jamming, spoofing, and cyber-attacks, and review countermeasures including secure routing and intrusion detection systems [102]. In parallel, low-altitude economy research highlights the importance of robust communication and traffic management for safe large-scale UAV operations in dense airspace [104].

Finally, swarm intelligence principles—drawing inspiration from biological systems such as insect colonies and bird flocks—have bridged network science with decentralized UAV coordination. These approaches leverage local interaction rules to achieve global robustness and adaptability [105]. Recent algorithms incorporate virtual leaders, attention mechanisms, and distributed optimization for tasks such as patrolling, tracking, and coverage [106]. Deep reinforcement learning methods further extend these capabilities, enabling the synthesis of distributed swarm policies directly from interaction data [99].

In summary, existing research on multi-UAV cooperative transport has made significant strides in modeling, formation control, trajectory planning, and admittance-based strategies, with growing integration of network science principles to enhance scalability and robustness. However, most studies remain constrained by several critical limitations as shown in Table I: heavy reliance on indoor motion capture systems, validation with payloads significantly below single-UAV capacity, and inadequate robustness against real-world uncertainties such as thrust saturation, cable-length variations, and communication failures. These gaps highlight the need for a distributed, computationally efficient framework that ensures stability under heavy-load conditions, accommodates system heterogeneity, and

operates reliably in outdoor environments—addressing the core challenges of practical deployment in the low-altitude economy.

III. PRELIMINARY AND PROBLEM FORMULATION

A. Model of the UAV-Cable-Load Transport System

In the cooperative transport scene, the UAV-cable-load system comprises n UAVs and a single load in $\mathbb{R}^3$. Each UAV is tethered to the load via a cable, as illustrated in Fig. 2. The dynamics of the i th UAV are modeled as a point-mass system:

$$\begin{cases} \dot{\mathbf{p}}_i = \dot{\mathbf{v}}_i \\ \dot{\mathbf{v}}_i = \frac{1}{m_i} (\mathbf{T}_i + \mathbf{G}_i - \mathbf{f}_i), \end{cases} \quad (1)$$

where $\mathbf{p}_i = [x_i \ y_i \ z_i]^T \in \mathbb{R}^3$ and $\mathbf{v}_i \in \mathbb{R}^3$ represent the position and velocity of the i th UAV, respectively; m_i denotes its mass, and $\mathbf{T}_i$, $\mathbf{G}_i$, and $\mathbf{f}_i$ are the propulsive force input, gravitational force, and cable tension acting on the UAV, respectively, for $i = 1, 2, \dots, n$.

The dynamics of the load are also described by a double-integrator model:

$$\begin{cases} \dot{\mathbf{p}}_1 = \mathbf{v}_1 \\ \dot{\mathbf{v}}_1 = \mathbf{G}_1 + \mathbf{f}_1, \end{cases} \quad (2)$$

where $\mathbf{p}_1 \in \mathbb{R}^3$ and $\mathbf{v}_1 \in \mathbb{R}^3$ denote the position and velocity of the load; m_1 is the load's mass; and $\mathbf{f}_1$ is the resultant force acting on the load. The relation between $\mathbf{f}_1$ and $\mathbf{f}_i$ is given by:

$$\mathbf{f}_1 = \sum_{i=1}^n \mathbf{f}_i, \quad i = 1, 2, \dots, n. \quad (3)$$

The cable force model is expressed as:

$$\begin{cases} \mathbf{f}_i = \max(\min(A_i, f_{\max,i}), 0) (-\mathbf{t}_{i0}) \\ A_i = k_i (l_{i1} - l_{i10}) + b_i dl_{i1} \\ dl_{i1} = d\mathbf{t}_i \cdot \mathbf{t}_{i0} \\ \mathbf{t}_i = \mathbf{p}_1 - \mathbf{p}_i \\ \mathbf{t}_{i0} = \frac{\mathbf{t}_i}{\|\mathbf{t}_i\|} \\ l_{i1} = \|\mathbf{t}_i\|, \end{cases} \quad (4)$$

where A_i denotes the cable tension generated by its elastic deformation, $f_{\max,i}$ is the maximum allowable tension before cable failure, $\mathbf{t}_i$ is the vector from the i th UAV to the load, and $\mathbf{t}_{i0}$ is the corresponding unit vector. Parameters k_i and b_i represent the stiffness and damping coefficients of the cable, respectively. l_{i1} and l_{i10} are the current and rest lengths of the cable. The model in (4) assumes the cable is massless, only exerts tensile force (no compression), allows minor elastic deformations, and has an upper tension threshold [55].

B. Objective

Given the UAV-cable-load system described in (1)–(4), the objective is to design a distributed controller that generates the propulsive force input $\mathbf{T}_i$ for each UAV, such that the load velocity $\mathbf{v}_1$ asymptotically converges to the desired velocity $\mathbf{v}_d$.

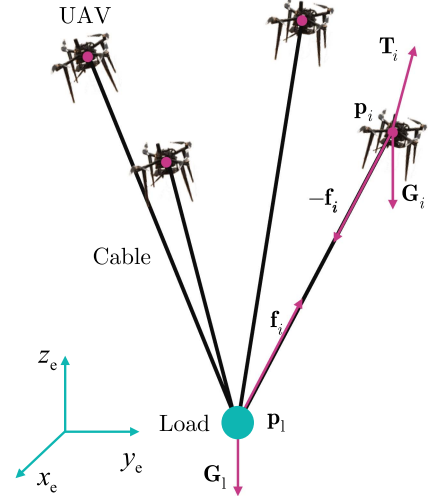


Fig. 2. The UAV-cable-load system. It comprises n UAVs and a single load. Each UAV is tethered to the load via a cable. The North-East-Down (NED) coordinate system is employed in this work for describing system variables. Each UAV is subject to rotor thrust, cable tension, and gravity. The payload experiences the combined tension from all cables in addition to its own weight.

IV. CONTROLLER DESIGN

A. Virtual Passive System Establishment

A virtual passive system is established for the UAV-cable-load configuration, incorporating the *virtual tray*, *virtual nodes*, and *virtual springs*, as illustrated in Fig. 3. The coordinates of the center of the virtual tray are defined as $\mathbf{q}_u = [x_u \ y_u \ z_u]^T \in \mathbb{R}^3$, and the virtual node corresponding to the load is $\mathbf{q}_1 = [x_1 \ y_1 \ z_1]^T \in \mathbb{R}^3$. The coordinates of the i th UAV's corresponding virtual node are denoted as $\mathbf{q}_i$, satisfying the following relationship [55]:

$$\mathbf{q}_i = \mathbf{p}_1 + \frac{|z_u - z_1|}{|z_1 - z_i|} (\mathbf{p}_i - \mathbf{p}_1), \quad i = 1, 2, \dots, n. \quad (5)$$

In this virtual passive system, the virtual tray is modeled as a sufficiently large yet finite disk. All virtual nodes corresponding to the UAVs are positioned on this virtual tray—i.e., they share the same height as the tray's center—but are allowed to slide across its surface ($\mathbf{q}_i = [\bar{x}_i \ \bar{y}_i \ \bar{z}_i]^T = [\bar{x}_i \ \bar{y}_i \ z_u]^T$). All virtual nodes are interconnected by virtual springs, which also connect each node to the center of the virtual tray. [48]. The virtual passive system is initialized based on the physical UAV-cable-load system's initial state. Specifically, the initial positions of the virtual nodes $\mathbf{q}_i$ (for $i = 1, 2, \dots, n$) are computed using (5), with the initial UAV positions $\mathbf{p}_i(0)$ and load position $\mathbf{p}_1(0)$ as inputs. This ensures that the virtual nodes align with the physical system's geometry at startup.

The dynamics of this virtual passive system are governed by the following Lagrangian equation:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{q}}_i} - \frac{\partial L}{\partial \mathbf{q}_i} + \frac{\partial \Psi}{\partial \dot{\mathbf{q}}_i} = \mathbf{U}_i, \quad i = 1, 2, \dots, n, u, l. \quad (6)$$

Here, L denotes the Lagrangian function of the system, Ψ is the dissipation function, and $\mathbf{U}_i$ represents the external driving

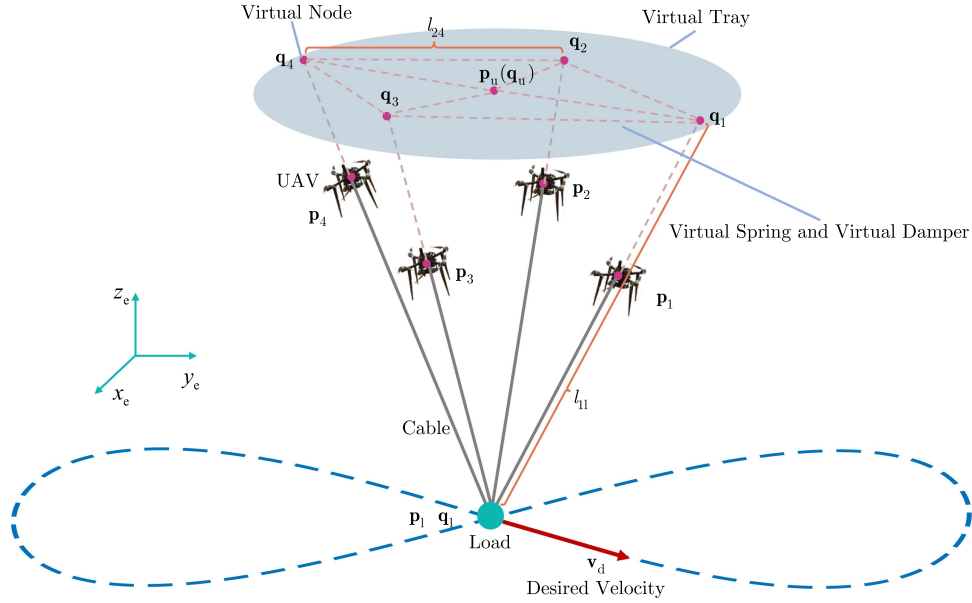


Fig. 3. An example of a virtual passive system for the UAV-cable-load system. This virtual passive system comprises three key components: a virtual tray, virtual nodes, and virtual spring-dampers. The blue ellipse represents the virtual tray, with pink points on its perimeter denoting the virtual nodes. Virtual springs and virtual dampers are depicted by pink dashed lines connecting adjacent nodes and gray dashed lines linking nodes to their corresponding UAVs. A blue dashed lemniscate curve shows the desired trajectory, while a red arrow indicates the load's current target velocity.

force acting on the intermediate virtual system [107], [108]. $i = u$ denotes the virtual node corresponding to the virtual tray, $i = l$ denotes the load. The Lagrangian function is defined as:

$$L = T - V = T - (V_G + V_E), \quad (7)$$

where T is the kinetic energy, V_G is the gravitational potential energy, and V_E denotes the elastic potential energy stored in the virtual springs.

The kinetic energy T is the sum of the kinetic energies of all virtual nodes and the load:

$$T = \sum_{i=1}^{n+2} \frac{1}{2} m_i \|\dot{\mathbf{q}}_i\|^2, \quad (8)$$

where m_i denotes the mass associated with each virtual node, and m_l is the mass of the load. The cumulative upperbound is set to $n + 2$ because this accounts for the virtual tray node $i = u$ and the virtual node corresponding to the load $i = l$, in addition to the n virtual nodes corresponding to the UAVs.

The gravitational potential energy V_G is given by

$$V_G = \sum_{i=1}^{n+2} m_i \mathbf{q}_i^T \mathbf{g}, \quad \mathbf{g} = [0 \ 0 \ g]^T, \quad (9)$$

where g denotes the gravitational acceleration.

The elastic potential energy V_E is formulated as

$$V_E = \frac{1}{4} \sum_{i,j=1}^{n+2} w_{i,j} k_{i,j} \Delta \bar{l}_{ij}^2, \quad (10)$$

where $\Delta \bar{l}_{ij} = \bar{l}_{ij0} - \bar{l}_{ij}$ represents the change in length of the virtual spring relative to its original length, $k_{i,j} > 0$ denotes the elastic coefficient of the virtual spring between the i th and j th

nodes, and $\mathbf{W} = [w_{ij}]_{(n+2) \times (n+2)}$ is the connection matrix. In this matrix, $w_{ij} = 1$ indicates that the i th and j th virtual nodes are connected by a spring. A typical example of $\mathbf{W}$ is:

$$\mathbf{W} = \begin{bmatrix} 0 & 1 & 1 & \cdots & 1 & 1 \\ 1 & 0 & 1 & \cdots & 1 & 1 \\ 1 & 1 & 0 & \cdots & 1 & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & 1 & \cdots & 0 & 0 \\ 1 & 1 & 1 & \cdots & 0 & 0 \end{bmatrix}_{(n+2) \times (n+2)}.$$

The variable $\bar{l}_{ij}$ denotes the current length of the virtual spring between nodes i and j , while $\bar{l}_{ij0}$ is the corresponding initial length. The current length is defined as:

$$\bar{l}_{ij} = \sqrt{\|\mathbf{q}_i - \mathbf{q}_j\|^2}, \quad i, j = 1, 2, \dots, n, u, l, i \neq j. \quad (11)$$

Meanwhile, the initial spring length $\bar{l}_{ij0}$ can be set according to the desired formation of the UAV-cable-load system or based on the pre-takeoff positions $\mathbf{q}_{i0}$ of the UAVs and other geometric constraints, such as cable lengths:

$$\bar{l}_{ij0} = \sqrt{\|\mathbf{q}_{i0} - \mathbf{q}_{j0}\|^2}, \quad i, j = 1, 2, \dots, n, u, l, i \neq j. \quad (12)$$

To account for energy dissipation within the virtual passive system—a crucial aspect for ensuring asymptotic stability and modeling realistic damping effects—we introduce a dissipation function Ψ . This function captures the non-conservative forces arising from virtual dampers and friction, which are essential for shaping the system's transient response and guaranteeing stability. Any force that leads to energy dissipation as the system evolves can be classified as a dissipative force. The dissipation

function can be derived by integrating the generalized velocity with the corresponding dissipative force [57], [109].

In this paper, the dissipation function Ψ is composed of two components:

$$\Psi = \Psi_d + \Psi_f, \quad (13)$$

where Ψ_d accounts for energy dissipation due to virtual dampers between virtual nodes, and Ψ_f represents energy loss due to friction between the virtual nodes and the medium through which they move.

These dissipation functions are formulated as follows:

$$\begin{cases} \Psi_d = \frac{1}{2} c_1 \sum_{i,j=1}^{n+2} \left(\dot{\Delta l}_{ij} \right)^2 \\ \Psi_f = \frac{1}{2} c_2 \sum_{i=1}^{n+2} \dot{\mathbf{q}}_i^T \dot{\mathbf{q}}_i \end{cases}, \quad (14)$$

where $c_1 > 0$ is the damping coefficient of the virtual dampers, and $c_2 > 0$ is the friction coefficient between the virtual nodes and their surrounding medium.

The generalized Lagrangian coordinates are defined as:

$$\mathbf{q} = [\mathbf{q}_1^T \quad \mathbf{q}_2^T \quad \cdots \quad \mathbf{q}_{n+2}^T]^T. \quad (15)$$

Based on (6)–(15), the dynamics of the virtual passive system can be expressed as:

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{G} + \mathbf{F}_{s(\mathbf{q})} + \mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} + \mathbf{F}_{f(\dot{\mathbf{q}})} = \mathbf{U}, \quad (16)$$

where $\mathbf{M} \in \mathbb{R}^{(3n+6) \times (3n+6)}$ is the mass matrix of the virtual passive system, given by:

$$\mathbf{M} = \text{diag}(m_1, m_1, m_1, m_2, m_2, m_2, \dots, m_{n+2}, m_{n+2}, m_{n+2}).$$

The gravitational force is given by:

$$\mathbf{G} = \mathbf{M} [\mathbf{g}^T \quad \mathbf{g}^T \quad \cdots \quad \mathbf{g}^T]^T. \quad (17)$$

The remaining terms in (16) represent: $\mathbf{F}_{s(\mathbf{q})}$ represents elastic forces due to the virtual springs, $\mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})}$ represents damping forces due to virtual dampers, $\mathbf{F}_{f(\dot{\mathbf{q}})}$ represents friction forces due to medium interaction. The elastic force term is:

$$\mathbf{F}_{s(\mathbf{q})} = \begin{bmatrix} \sum_{j \in \mathcal{N}_{n+2,1}} k_{1j} w_{1j} \left(1 - \frac{\bar{l}_{1j0}}{l_{1j}} \right) (\mathbf{q}_1 - \mathbf{q}_j) \\ \sum_{j \in \mathcal{N}_{n+2,2}} k_{2j} w_{2j} \left(1 - \frac{\bar{l}_{2j0}}{l_{2j}} \right) (\mathbf{q}_2 - \mathbf{q}_j) \\ \vdots \\ \sum_{j \in \mathcal{N}_{n+2,1}} k_{lj} w_{lj} \left(1 - \frac{\bar{l}_{lj0}}{l_{lj}} \right) (\mathbf{q}_l - \mathbf{q}_j) \end{bmatrix}. \quad (18)$$

The damping force term is:

$$\mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} = \begin{bmatrix} c_1 \sum_{j \in \mathcal{N}_{n+2,1}} w_{1j} (\dot{\mathbf{q}}_1 - \dot{\mathbf{q}}_j) \\ c_1 \sum_{j \in \mathcal{N}_{n+2,2}} w_{2j} (\dot{\mathbf{q}}_2 - \dot{\mathbf{q}}_j) \\ \vdots \\ c_1 \sum_{j \in \mathcal{N}_{n+2,1}} w_{lj} (\dot{\mathbf{q}}_l - \dot{\mathbf{q}}_j) \end{bmatrix}. \quad (19)$$

The friction force term is:

$$\mathbf{F}_{f(\dot{\mathbf{q}})} = \begin{bmatrix} c_2 \dot{\mathbf{q}}_1 \\ c_2 \dot{\mathbf{q}}_2 \\ \vdots \\ c_2 \dot{\mathbf{q}}_l \end{bmatrix}. \quad (20)$$

Here, $\mathcal{N}_{m,i} = \{m = 1, 2, \dots, n, u, l, m \neq i\}$ denotes the set of indices of all virtual nodes connected to the i th node.

The established virtual passive system inherently constitutes a dynamic networked control system. Each UAV, along with its associated virtual node, acts as a dynamical agent in the network. The virtual springs and dampers define the coupling links between these agents, forming a strongly connected physical interaction topology. The distributed control law, as will be presented, requires only states from locally coupled neighbors, adhering to the principle of networked control systems. This architecture ensures scalability and resilience against single-point failures, which are critical requirements for large-scale UAV operations in low-altitude economy.

Proposition 1: The virtual passive system described by (16) is output strictly passive.

Proof: Consider the virtual passive system with output selected as $\mathbf{y} = \dot{\mathbf{q}}$. The state-space form of the system can be written as:

$$\ddot{\mathbf{q}} = \mathbf{M}^{-1} (-\mathbf{G} - \mathbf{F}_{s(\mathbf{q})} - \mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} - \mathbf{F}_{f(\dot{\mathbf{q}})}) + \mathbf{M}^{-1} \mathbf{U}. \quad (21)$$

Define the memoryless energy storage (Lyapunov-like) function as:

$$S(\mathbf{q}) = T + V = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} + V_{G(\mathbf{q})} + V_{E(\mathbf{q})} \geq 0, \quad (22)$$

which includes both kinetic and potential energy terms.

The term “memoryless energy storage function” refers to a Lyapunov-like function that depends only on the current system state without incorporating historical states or integral terms. This is in contrast to storage functions that may include integral terms or other memory elements [56], [57]. Taking the time derivative of (22) yields:

$$\begin{aligned} \dot{S}(\mathbf{q}) &= \dot{\mathbf{q}}^T \mathbf{M} \ddot{\mathbf{q}} + \dot{\mathbf{q}}^T (\mathbf{G} + \mathbf{F}_{s(\mathbf{q})}) \\ &= \dot{\mathbf{q}}^T (\mathbf{U} - \mathbf{G} - \mathbf{F}_{s(\mathbf{q})} - \mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} - \mathbf{F}_{f(\dot{\mathbf{q}})}) \\ &\quad + \dot{\mathbf{q}}^T (\mathbf{G} + \mathbf{F}_{s(\mathbf{q})}) \\ &= \dot{\mathbf{q}}^T (\mathbf{U} - \mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} - \mathbf{F}_{f(\dot{\mathbf{q}})}). \end{aligned} \quad (23)$$

Substituting $\mathbf{y} = \dot{\mathbf{q}}$, we can express the derivative as:

$$\begin{aligned} \dot{S}(\mathbf{q}) &= \mathbf{y}^T \mathbf{U} - \mathbf{y}^T (\mathbf{F}_{d(\mathbf{q},\dot{\mathbf{q}})} + \mathbf{F}_{f(\dot{\mathbf{q}})}) \\ &= \mathbf{U}^T \mathbf{y} - c_1 \sum_{i,j=1}^{n+2} \frac{w_{ij}}{l_{ij}^2} [(\dot{\mathbf{q}}_i - \dot{\mathbf{q}}_j)^T (\mathbf{q}_i - \mathbf{q}_j)]^2 \\ &\quad - c_2 \sum_{i=1}^{n+2} \dot{\mathbf{q}}_i^T \dot{\mathbf{q}}_i \\ &\leq \mathbf{U}^T \mathbf{y} - c_2 \mathbf{y}^T \mathbf{y}, \end{aligned} \quad (24)$$

Since the system satisfies the *strict output passivity inequality*, the virtual passive system is output strictly passive according to **Definition 2.1** of [57]. $\square$

Corollary 1: The virtual passive system (16) exhibits asymptotic stability under zero input [56], [57].

Proof: When $\mathbf{U} = \mathbf{0}$, according to (24), it follows that

$$\dot{S}_{(\mathbf{q})} \leq -c_2 \mathbf{y}^T \mathbf{y} \leq 0, \quad (25)$$

with equality if and only if $\mathbf{y} = \dot{\mathbf{q}} = \mathbf{0}$. By LaSalle's Invariance Principle, since $S_{(\mathbf{q})}$ is radially unbounded (i.e., $S_{(\mathbf{q})} \rightarrow \infty$ as $\mathbf{q} \rightarrow \infty$), and in light of (25), all system trajectories converge to the invariant set

$$\mathcal{M} = \{\dot{\mathbf{q}} \mid \dot{\mathbf{q}} = \mathbf{0}\}. \quad (26)$$

Physically, this means that in the absence of external forces, the internal virtual system will always converge to a completely stationary and stable state, regardless of initial conditions.

Corollary 2: The virtual passive system (16) satisfies finite-gain $\mathcal{L}_2$ stability [56], [57]. $\square$

Proof: The time derivative of the energy storage function of the virtual passive system satisfies

$$\begin{aligned} \dot{S}_{(\mathbf{q})} &\leq \mathbf{U}^T \mathbf{y} - c_2 \mathbf{y}^T \mathbf{y} \\ &= -\frac{1}{2c_2} (\mathbf{U} - c_2 \mathbf{y})^T (\mathbf{U} - c_2 \mathbf{y}) + \frac{1}{2c_2} \mathbf{U}^T \mathbf{U} - \frac{c_2}{2} \mathbf{y}^T \mathbf{y} \\ &\leq \frac{1}{2c_2} \mathbf{U}^T \mathbf{U} - \frac{c_2}{2} \mathbf{y}^T \mathbf{y}. \end{aligned} \quad (27)$$

Integrating both sides of (27) with respect to time yields

$$\begin{aligned} \int_0^t \mathbf{y}^T(\tau) \mathbf{y}(\tau) d\tau &\leq \frac{1}{c_2^2} \int_0^t \mathbf{U}^T(\tau) \mathbf{U}(\tau) d\tau \\ &\quad - \frac{2}{c_2} (S_{(\mathbf{q}(t))} - S_{(\mathbf{q}(0))}). \end{aligned} \quad (28)$$

Since $S_{(\mathbf{q}(t))} \geq 0$, we have

$$\int_0^t \mathbf{y}^T(\tau) \mathbf{y}(\tau) d\tau \leq \frac{1}{c_2^2} \int_0^t \mathbf{U}^T(\tau) \mathbf{U}(\tau) d\tau + \frac{2}{c_2} S_{(\mathbf{q}(0))}, \quad (29)$$

which leads to

$$\|\mathbf{y}(t)\|_{\mathcal{L}_2}^2 \leq \frac{1}{c_2^2} \|\mathbf{U}(t)\|_{\mathcal{L}_2}^2 + \frac{2}{c_2} S_{(\mathbf{q}(0))}. \quad (30)$$

Thus,

$$\|\mathbf{y}(t)\|_{\mathcal{L}_2} \leq \frac{1}{c_2} \|\mathbf{U}(t)\|_{\mathcal{L}_2} + \sqrt{\frac{2}{c_2} S_{(\mathbf{q}(0))}} = \gamma \|\mathbf{U}(t)\|_{\mathcal{L}_2} + \beta, \quad (31)$$

where $\gamma = \frac{1}{c_2}$ is the gain, and $\beta = \sqrt{\frac{2}{c_2} S_{(\mathbf{q}(0))}}$ is the bias.

Hence, the virtual passive system is finite-gain $\mathcal{L}_2$ -stable, indicating that the system output remains bounded for any input with finite energy. $\square$

B. Velocity Tracking Controller Design

The following section focuses on the velocity tracking control problem of the virtual passive system. The objective is to design

an appropriate control input $\mathbf{U}$ such that the output $\mathbf{y} = \dot{\mathbf{q}} = \mathbf{v}$ of the virtual passive system (16) can accurately track a bounded desired output $\mathbf{v}_d(t)$ (with $\dot{\mathbf{v}}_d(t)$ also being bounded).

For the i th virtual node, the dynamics governed by (16) can be expressed as:

$$\begin{aligned} m_i \dot{\mathbf{v}} + m_i \mathbf{g} + \mathbf{F}_{s(\mathbf{q}_i)} + \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} + \mathbf{F}_{f(\dot{\mathbf{q}}_i)} &= \mathbf{u}_i, \\ \mathbf{F}_{s(\mathbf{q}_i)} &= \sum_{j \in N_{n+2,i}} k_{ij} w_{ij} \left(1 - \frac{\bar{l}_{ij0}}{\bar{l}_{ij}}\right) (\mathbf{q}_i - \mathbf{q}_j), \\ \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} &= c_1 \sum_{j \in N_{n+2,i}} w_{ij} (\dot{\mathbf{q}}_i - \dot{\mathbf{q}}_j), \\ \mathbf{F}_{f(\dot{\mathbf{q}}_i)} &= c_2 \dot{\mathbf{q}}_i, \\ i &= 1, 2, \dots, n+2. \end{aligned} \quad (32)$$

Let the tracking error of the i th virtual node be defined as

$$\mathbf{e}_i(t) = \mathbf{v}_i(t) - \mathbf{v}_{id}(t), \quad (33)$$

where the desired velocity satisfies $\mathbf{v}_{id} = \mathbf{v}_d$. This implies that we aim for the entire virtual passive system to track a common desired velocity $\mathbf{v}_d$, rather than requiring each virtual node to follow its own individual desired velocity.

The control input is designed as follows:

$$\mathbf{u}_i = \underbrace{(m_i \dot{\mathbf{v}}_{id}(t) + m_i \mathbf{g} + c_2 \dot{\mathbf{q}}_{id}(t))}_{\mathbf{u}_{ff i}} - \underbrace{\mathbf{K} \mathbf{e}_i(t)}_{\mathbf{u}_{fb i}}, \quad (34)$$

where $\mathbf{u}_{ff}$ serves as the feed forward term designed to compensate for the system dynamics, while $\mathbf{u}_{fb}$ is the feedback term, and $\mathbf{K}$ is a positive definite matrix.

Proposition 2: Suppose that the i th virtual node's control input is written as in (34), then the output of the virtual passive system (32), $\mathbf{y} = \dot{\mathbf{q}} = \mathbf{v}$, asymptotically tracks the given reference velocity signal $\mathbf{v}_d(t)$ (with $\dot{\mathbf{v}}_d(t)$ also being bounded). That is, we have:

$$\lim_{t \rightarrow \infty} \|\mathbf{e}_i(t)\| = \lim_{t \rightarrow \infty} \|\mathbf{v}_i(t) - \mathbf{v}_{id}(t)\| = 0. \quad (35)$$

Proof: See Appendix A.

Remark 1: If one wishes to reduce the dependence of the control input on model information—such as omitting compensation for gravitational force—the control input can be redesigned as:

$$\mathbf{u}_i = \underbrace{(m_i \dot{\mathbf{v}}_{id}(t) + c_2 \dot{\mathbf{q}}_{id}(t))}_{\mathbf{u}_{ff i}} - \underbrace{\mathbf{K} \mathbf{e}_i(t)}_{\mathbf{u}_{fb i}}. \quad (36)$$

If the Lyapunov function is simply constructed in the style of a memoryless energy storage function (as in (22)), we define:

$$\begin{aligned} V_S &= T(\mathbf{e}_i) + V_E + V_G \\ &= \frac{1}{2} \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \mathbf{e}_i + \frac{1}{4} \sum_{i,j=1}^{n+2} \Delta \bar{l}_{ij}^2 + \sum_{i=1}^{n+2} m_i \mathbf{q}_i^T \mathbf{g}. \end{aligned} \quad (37)$$

Taking the time derivative of the Lyapunov function (37) yields:

$$\dot{V}_S = \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \dot{\mathbf{e}}_i + \frac{1}{2} \sum_{i,j=1}^{n+2} \Delta \bar{l}_{ij} \Delta \dot{\bar{l}}_{ij} + \sum_{i=1}^{n+2} m_i \mathbf{v}_i^T \mathbf{g}. \quad (38)$$

Considering the dynamics with omitted gravity compensation, we have:

$$\begin{aligned} \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \dot{\mathbf{e}}_i &= \sum_{i=1}^{n+2} \mathbf{e}_i^T [-m_i \mathbf{g} - \mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} \\ &\quad - (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i] \\ &= \sum_{i=1}^{n+2} \mathbf{e}_i^T (-\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)}) - \sum_{i=1}^{n+2} \mathbf{e}_i^T m_i \mathbf{g} \\ &\quad - \sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i. \end{aligned} \quad (39)$$

Following an argument similar to that in Proposition 2, we obtain:

$$\begin{aligned} \dot{V}_S &\leq - \sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i - \sum_{i=1}^{n+2} m_i \mathbf{v}_d^T \mathbf{g} \\ &\quad - \frac{1}{2} \sum_{i,j=1}^{n+2} c_1 w_{ij} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{v}_i - \mathbf{v}_j). \end{aligned} \quad (40)$$

If the desired velocity $\mathbf{v}_d$ is perpendicular to the gravitational vector, i.e., $\mathbf{v}_d \perp \mathbf{g}$, the term $\mathbf{v}_d^T \mathbf{g} = 0$, and the Lyapunov function derivative remains negative semi-definite. Thus, a conclusion similar to Proposition 2 holds: the system can still asymptotically track the desired velocity.

Physical interpretation: When the desired velocity has no component in the direction of gravity, even without explicitly compensating for gravitational forces in the control law, the system is still capable of asymptotically following the desired velocity.

However, if $\mathbf{v}_d^T \mathbf{g} \neq 0$, the system output remains ultimately uniformly bounded. While the system state does not diverge, the tracking error does not converge to zero, and a steady-state error persists. When the tracking error $\mathbf{e}_i$ is large, the negative-definite term $-\sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i$ dominates, driving the error to diminish. Eventually, the system enters a bounded set M , defined as:

$$M = \left\{ \mathbf{e}_i \mid \sum_{i=1}^{n+2} \mathbf{e}_i^T \mathbf{e}_i \leq \frac{\left| \sum_{i=1}^{n+2} m_i \mathbf{v}_d^T \mathbf{g} \right|}{\lambda_{\min}(c_2 \mathbf{I} + \mathbf{K})} \right\}. \quad (41)$$

Therefore, instead of directly using the energy storage memoryless function, the Lyapunov function can be refined as:

$$\begin{aligned} V_S &= T(\mathbf{e}_i) + V_E + \tilde{V}_G \\ &= \frac{1}{2} \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \mathbf{e}_i + \frac{1}{4} \sum_{i,j=1}^{n+2} \Delta \bar{l}_{ij}^2 + \sum_{i,j=1}^{n+2} m_i (\mathbf{q}_i - \mathbf{q}_{id})^T \mathbf{g}, \end{aligned} \quad (42)$$

where $\dot{\mathbf{q}}_{id} = \mathbf{v}_d$. In this case, the derivative of the Lyapunov function (42) satisfies:

$$\begin{aligned} \dot{V}_S &= - \sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i \\ &\quad - \frac{1}{2} \sum_{i,j=1}^{n+2} c_1 w_{ij} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{v}_i - \mathbf{v}_j) \leq 0. \end{aligned} \quad (43)$$

Similar to the conclusion in Proposition 2, all system trajectories converge to the invariant set (59). As a result, the virtual passive system can asymptotically track the desired velocity $\mathbf{v}_d$, while the individual virtual nodes within the system remain relatively stationary with respect to one another.

Remark 2: The theoretical analysis in Proposition 2 assumes the desired velocity $\mathbf{v}_d(t)$ is bounded and first-order differentiable to ensure the existence of $\dot{\mathbf{v}}_d(t)$ in the Lyapunov function derivative. In practical applications, velocity commands from a high-level planner (e.g., waypoint-based navigation) may be discontinuous. To handle such commands, a reference pre-filter or trajectory smoother (e.g., a first-order low-pass filter) can be employed to generate a smoothed, continuously differentiable $\mathbf{v}_d(t)$ that satisfies the theoretical assumptions. This preprocessing step is standard in motion control systems, maintains the stability guarantees of the controller, and prevents excitation of high-frequency dynamics. Thus, the proposed framework remains robust to practical input variations.

Based on the aforementioned theoretical analysis, the controller design for multi-UAV cooperative transportation-developed within the passive system framework proposed in this work-achieves the following: First, in the absence of external input, the dissipative forces within the virtual passive system drive the entire UAV-cable-load system to evolve into a stable and robust integration of virtual and physical dynamics. This integrated system demonstrates strong self-regulation and self-stabilization capabilities when subjected to disturbances, uncertainties, and state transitions, ultimately converging to a stationary and stable equilibrium. Second, building upon this foundation, a control input is designed for the virtual passive system to track a desired velocity. Internally, the virtual passive system attains a stable state in which the virtual nodes remain relatively stationary with respect to one another, while the system as a whole is capable of asymptotically following the target velocity. In summary, the proposed controller design method, grounded in passive system theory, enables a clear decoupling between internal coordination and overall trajectory tracking of the multi-agent hoisting system. This separation allows for the independent design of a velocity-tracking control law to steer the virtual system toward the desired motion, offering significant practical advantages for real-world implementation.

C. Implement

In practice, the design of the multi-UAV cooperative transportation controller based on the passive system can be carried out in two stages:

The first stage involves constructing a virtual passive system without any external input. Based on (5) and the actual states of the UAV-cable-load system, a virtual system is established, forming a hybrid system that couples the virtual model with the physical system, as illustrated in Fig. 3. For the dynamics of the virtual passive system:

$$m_i \ddot{\mathbf{q}}_i + m_i \mathbf{g} + \mathbf{F}_{s(\mathbf{q}_i)} + \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} + \mathbf{F}_{f(\dot{\mathbf{q}}_i)} = \mathbf{0}. \quad (44)$$

The controllable variable is $\ddot{\mathbf{q}}_i$. If $\ddot{\mathbf{q}}_i$ can track the desired $\ddot{\mathbf{q}}_{id}$:

$$\ddot{\mathbf{q}}_{id} = -\mathbf{g} - \frac{\mathbf{F}_{s(\mathbf{q}_i)} + \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} + \mathbf{F}_{f(\dot{\mathbf{q}}_i)}}{m_i}, \quad (45)$$

then the virtual system becomes a strictly output passive system, exhibiting the zero-input asymptotic stability and finite-gain $\mathcal{L}_2$ stability properties as established in Proposition 1, Corollaries 1, and 2.

The second stage involves designing a velocity-tracking controller for the entire virtual passive system, as described by (36), where $\mathbf{u}_i$ denotes the external force applied to the i th virtual node. In practice, the term $m_i \mathbf{g} + c_2 \dot{\mathbf{q}}_{id}$ can be treated as a disturbance. To mitigate steady-state error and improve regulation response, an integral-derivative feedback mechanism is incorporated into the control law:

$$\mathbf{u}_i = \underbrace{m_i \dot{\mathbf{v}}_d}_{\mathbf{u}_{ff i}} - \underbrace{\left(\mathbf{K}_p \mathbf{e}_i + \mathbf{K}_I \int \mathbf{e}_i dt + \mathbf{K}_d \frac{d}{dt} \mathbf{e}_i \right)}_{\mathbf{u}_{fb i}}. \quad (46)$$

Here, $\mathbf{e}_i = \dot{\mathbf{q}}_i - \mathbf{v}_d$ represents the velocity tracking error of the i th virtual node. The control input consists of a feed forward term $\mathbf{u}_{ff i}$ to match the desired acceleration, and a feedback term $\mathbf{u}_{fb i}$ to suppress tracking errors.

In line with Proposition 2. and Remark 2., this controller enables the virtual passive system to track the desired velocity $\mathbf{v}_d$ using $\mathbf{u}_i$. Under this control input, the acceleration induced in the i th virtual node is given by:

$$\ddot{\mathbf{q}}_{iu} = \dot{\mathbf{v}}_d - \frac{1}{m_i} \left(\mathbf{K}_p \mathbf{e}_i + \mathbf{K}_I \int \mathbf{e}_i dt + \mathbf{K}_d \frac{d}{dt} \mathbf{e}_i \right). \quad (47)$$

By integrating Steps One and Two, and mapping the desired acceleration of the virtual node back to the actual system via the geometric relationship in (5), the desired acceleration of the i th UAV is given by:

$$\mathbf{a}_{id} = \ddot{\mathbf{q}}_l + \frac{|z_l - z_i|}{|z_u - z_l|} (\ddot{\mathbf{q}}_{id} + \ddot{\mathbf{q}}_{iu} - \ddot{\mathbf{q}}_l), i = 1, 2, \dots, n + 2. \quad (48)$$

Then, according to the dynamic model of the actual UAV system in (1), the desired propulsive force for the i th UAV is calculated as:

$$\mathbf{T}_{id} = m_i \mathbf{a}_{id} - \mathbf{G}_i + \mathbf{f}_i. \quad (49)$$

Here, $\mathbf{f}_i$ denotes the main external disturbance acting on the UAV during the transportation process. It can be measured using a force sensor or estimated via a disturbance observer, such as the Extended State Observer (ESO), Compensation Function Observer (CFO), among others. In this paper, the CFO is employed to estimate $\mathbf{f}_i$.

In summary, the proposed controller achieves full decentralization and does not rely on direct communication between UAVs. Each UAV only requires relative position and velocity information of its neighbors, which can be acquired through relative localization techniques. This enables the system to function effectively even in communication-denied environments.

V. NUMERICAL SIMULATION

This section presents numerical simulation results to validate the performance and robustness of the proposed control method.

TABLE II
SYSTEM PARAMETERS AND CONTROL PARAMETERS IN THE SIMULATION

Parameter	Description	Values
n	Number of UAVs	10
m_i (kg)	Mass of UAVs	1
m_l (kg)	Mass of load	14
$f_{i,\max}$ (N)	Maximum load capacity	18
l_{il} (m)	Cable length	4.2, 4.7, 3.6, 4.3, 3.6, 3.4, 4.1, 3.9, 3.9, 4.2
k_{ij}	Elastic coefficient	7
c_1	Damping coefficient	1
c_2	Friction coefficient	5
$\mathbf{K}_p$	Proportional feedback gain	$\text{diag}(10, 10, 10)$
$\mathbf{K}_I$	Integral feedback gain	$\text{diag}(2, 2, 2)$
$\mathbf{K}_d$	Derivative feedback gain	$\text{diag}(0.1, 0.1, 0.1)$

The desired trajectory is set as a figure-eight path, with the corresponding desired velocity obtained by taking its time derivative:

$$x_d = 20 \sin\left(\frac{2\pi}{60}t\right), y_d = 10 \sin\left(\frac{4\pi}{60}t\right), z_d = 6.4,$$

$$v_{xd} = 20 \cdot \frac{2\pi}{60} \cos\left(\frac{2\pi}{60}t\right), v_{yd} = 10 \cdot \frac{4\pi}{60} \cos\left(\frac{4\pi}{60}t\right), v_{zd} = 0.$$

A video demonstration has been uploaded at https://youtu.be/L_SCzKD8-0~s, which briefly introduces the proposed method and visually presents the simulation results.

Up to ten UAVs are employed for cooperative transport, with their collective control enabling the load's velocity to follow the desired velocity profile. Each UAV is assumed to have a mass of $m_i = 1$ kg and a maximum thrust (load capacity) of 18 N. The load has a mass of $m_l = 14$ kg, which is relatively heavy compared to the total maximum thrust of 200 N from the ten UAVs. This highlights a key distinction of this work compared to previous studies, where significantly lighter loads were typically considered relative to the UAVs' total load capacity.

The cable lengths are provided in Table II, with each UAV connected to the load via a cable of distinct length. This demonstrates that the proposed control method does not require uniform cable lengths. Other parameters of the virtual passive system and controller settings are also listed in Table II.

As illustrated in Fig. 4, the ten UAVs collaboratively transport the load along the desired trajectory. From Fig. 4, the following observations can be made:

- 1) *Accurate Velocity Tracking*: The actual trajectory of the load closely follows the desired trajectory, demonstrating the effectiveness of the proposed cooperative velocity controller. (As this paper focuses on velocity tracking, a detailed analysis of trajectory tracking error is omitted.)
- 2) *Robustness to Cable Length Variations*: The visibly varying lengths of the UAV-load cables confirm the controller's capability to handle heterogeneity and uncertainties in cable lengths.
- 3) *Stability of the Virtual-Real System*: The spatial distribution of the virtual nodes, UAVs, and the load remains consistent and stable during motion, validating the quasi-static configuration predicted by theoretical analysis.

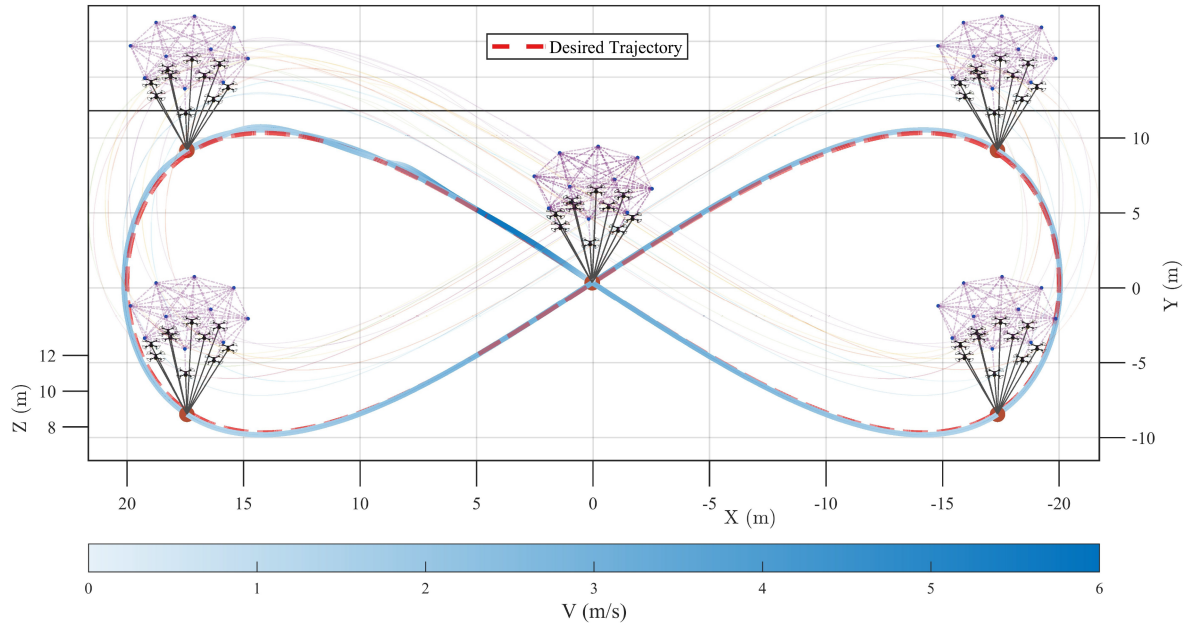


Fig. 4. Trajectories of UAVs and the load, distributions of virtual nodes and springs. The large red marker indicates the load's position. Blue dots represent the virtual nodes, while purple dashed lines depict virtual springs and dampers. Solid black lines represent the physical cables connecting UAVs to the load. The thick blue trajectory, color-mapped according to the load's velocity magnitude, indicates the actual load trajectory. The red dashed line represents the desired trajectory. The thinner semi-transparent lines illustrate the individual UAV trajectories.

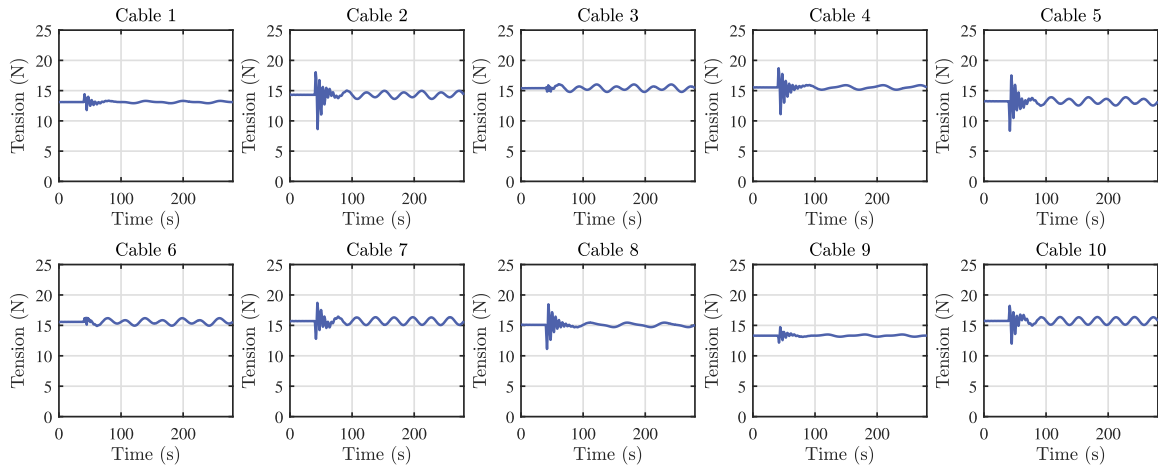


Fig. 5. Tension force of each cable.

Fig. 5 shows the tension forces in the ten cables, representing the load distribution among the UAVs. During the initial period from 0 s to 40 s, the system remains in an equilibrium state regulated by the virtual passive system. In this state, the tension forces are well-balanced, ranging between 13 N and 16 N. At 40 s, the UAV-cable-load system transitions from a standstill to tracking the desired velocity. As indicated in Fig. 4, the desired velocity reaches its peak value at 40 s. Consequently, the interval between 40 s and 60 s represents a dynamic transition phase. Although the cable tensions exhibit some transient fluctuations during this period, they remain well within the UAVs' maximum thrust limits. After 60 s, the system achieves stable velocity tracking, where the cable tensions oscillate periodically with

small amplitudes. The balanced load sharing throughout this phase indicates stable system performance.

Fig. 6 compares the desired and actual velocity components along the x and y axes, while Fig. 7 depicts the corresponding velocity tracking errors. The desired velocity starts at zero and steps to approximately 2.09 m/s at 40 s. The actual velocity quickly responds between 40 s and 50 s, effectively tracking the desired velocity. With the combined action of virtual springs, virtual dampers in the virtual-real passive system, and the proposed velocity tracking controller, the velocity tracking error converges rapidly. After 60 s, the tracking error remains negligible (less than 0.02 m/s), verifying the controller's effectiveness in achieving precise velocity tracking.

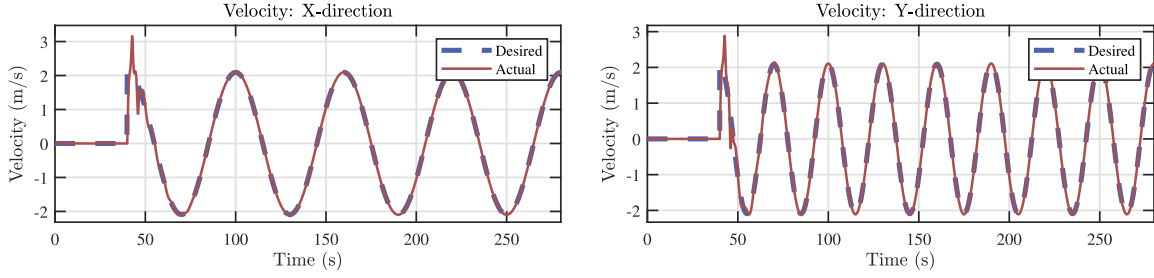


Fig. 6. Comparison between the desired velocity and the actual velocity.

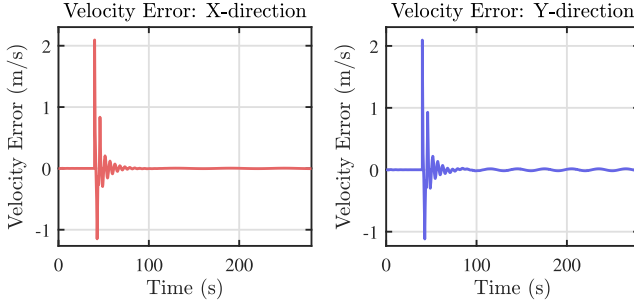


Fig. 7. Velocity tracking error.

TABLE III
SYSTEM PARAMETERS AND CONTROL PARAMETERS IN THE EXPERIMENTS

Parameter	Description	Values
n	Number of UAVs	6
m_i (kg)	Mass of UAVs	1.47
m_l (kg)	Mass of load	6
$f_{i,\max}$ (N)	Maximum load capacity	15.6
l_{il} (m)	Cable length	2
k_{ij}	Elastic coefficient	7
c_1	Damping coefficient	1
c_2	Friction coefficient	5
$\mathbf{K}_p$	Proportional feedback gain matrix	$\text{diag}(10, 10, 10)$
$\mathbf{K}_i$	Integral feedback gain matrix	$\text{diag}(2, 2, 2)$
$\mathbf{K}_d$	Derivative feedback gain matrix	$\text{diag}(0.1, 0.1, 0.1)$

Summary of Results: The simulation results validate that the proposed virtual-passivity-based velocity tracking controller enables cooperative aerial transportation of heavy loads using multiple UAVs, despite constraints such as limited thrust capacity, heterogeneous cable lengths, and significant load mass. The controller maintains safe inter-UAV distances, autonomously adjusts formation and individual load-sharing, and ensures system stability even under aggressive, step-changing inputs. The system rapidly converges to track the desired velocity profile, demonstrating strong robustness, scalability, and reliability.

VI. EXPERIMENTAL RESULTS

This section describes the hardware setup and presents the results of outdoor field experiments, which demonstrate the performance, effectiveness, and robustness of the proposed control method in real-world conditions.

Hardware Configuration: Up to six identical quadrotor UAVs were deployed in the outdoor experiments. Each UAV has a

mass of 1.47 kg and is capable of generating a maximum thrust of 30 N. A downward-facing fisheye camera is mounted on the load to capture its position. Each UAV is equipped with Differential GPS (DGPS) for accurate self-localization. Inter-UAV communication is established via the User Datagram Protocol (UDP), enabling the exchange of state information such as position and velocity. The proposed controller is implemented in a distributed manner, running independently on each UAV. In the current experimental setup, the relative positions between UAVs, required by the distributed controller, are obtained by sharing absolute DGPS coordinates via UDP communication. This implementation choice was made for experimental convenience and reliably provides the relative measurements needed by the control law. It is important to note that the proposed control framework is agnostic to the source of these relative measurements and is equally compatible with communication-free relative localization techniques, such as vision-based systems, highlighting its applicability in communication-denied environments.

Load and Experimental Setup: The transported load consists of multiple counterweights with a total mass of 6 kg, securely held within a cargo net. Tether cables connect the cargo net to the UAVs. Due to spatial constraints at the test site, two fixed locations were designated as reference points for the experiment. The desired velocity profile for the load is defined by (50).

$$\mathbf{v}_d = k(\mathbf{p}_d - \mathbf{p}_l), k = 1,$$

$$\mathbf{p}_d = \begin{cases} \mathbf{p}_{d1}, & t < 45 - \Delta t, \\ (1 - \phi(t))\mathbf{p}_{d1} + \phi(t)\mathbf{p}_{d2}, & 45 - \Delta t \leq t \leq 45 + \Delta t, \\ \mathbf{p}_{d2}, & t > 45 + \Delta t \end{cases}$$

$$\mathbf{p}_{d1} = [0 \ -16 \ 1]^T, \mathbf{p}_{d2} = [-1 \ -1 \ 1]^T,$$

$$\phi(t) = \frac{1}{1 + e^{-\sigma(t-45)}}, \sigma = 5. \quad (50)$$

Fig. 8 illustrates the trajectories of the load and UAVs during the outdoor experiment, along with the spatial distributions of the UAVs and virtual nodes at several representative time instants. The line styles and markers follow the same definitions as in Fig. 4. It is evident that, under the regulation of virtual nodes, springs, and dampers, the UAV-cable-load system maintained a stable formation throughout the cooperative transportation task, with the individual components exhibiting relatively quasi-static behavior.

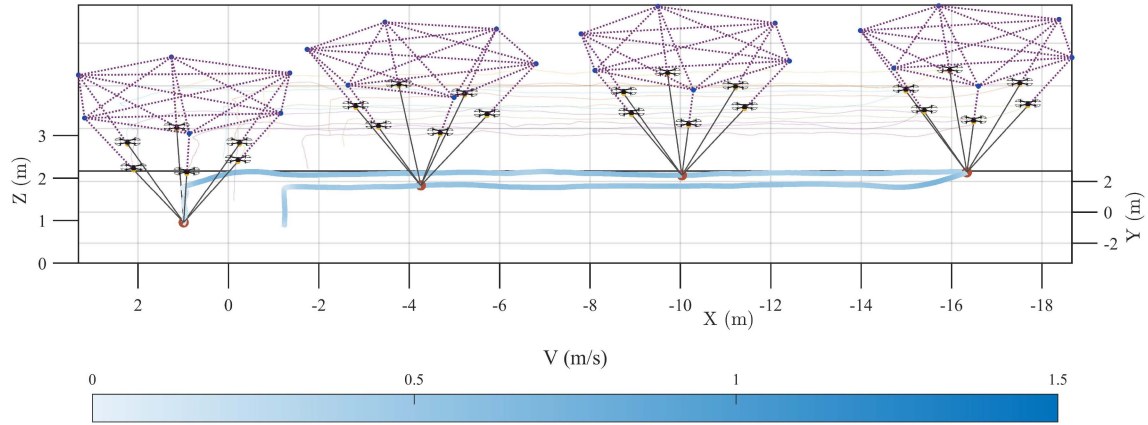


Fig. 8. Trajectories of UAVs and the load, distributions of virtual nodes and springs in the outdoor experiment.

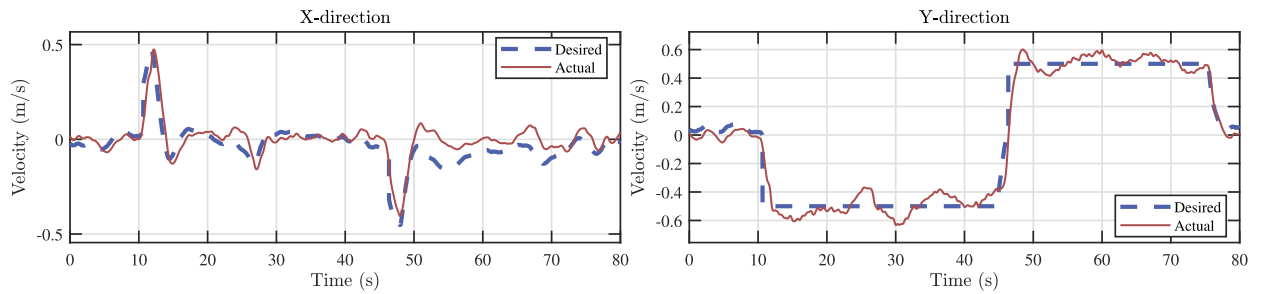


Fig. 9. Comparison between the desired velocity and the actual velocity in the outdoor experiment. Quantitative analysis of the velocity tracking performance during the stable flight phase ($t=20$ s to 40 s) shows an RMS error of 0.047 m/s and maximum error of 0.102 m/s, demonstrating the controller's accuracy in practical conditions.

Fig. 9 compares the desired and actual load velocities during the experiment. At $t = 10$ s, the UAV team completed takeoff and began tracking the reference velocity profile. The load reached the first target location at $t = 45$ s and subsequently proceeded toward the second target, following the desired velocity trajectory. It arrived and landed near the second target location at approximately $t = 80$ s. Despite observable fluctuations in the actual load velocity—primarily due to wind disturbances—the system effectively tracked the desired profile and completed the cooperative transportation task successfully.

Fig. 10 presents the estimated cable tension forces throughout the experiment. Significant tension variations are observed at $t = 10$ s (takeoff/transition) and $t = 45$ s (arrival at the first target), caused by dynamic maneuvers. Nevertheless, the load-sharing among the cables remained generally balanced across the transportation process.

In summary, the outdoor experimental results validate the effectiveness and robustness of the proposed virtual-passivity-based cooperative aerial transportation framework for multi-UAV systems operating in real-world environments.

VII. CONCLUSION

This paper proposed a virtual-passivity-based distributed control framework for cooperative heavy-load transportation using multiple UAVs under thrust limitations and heterogeneous

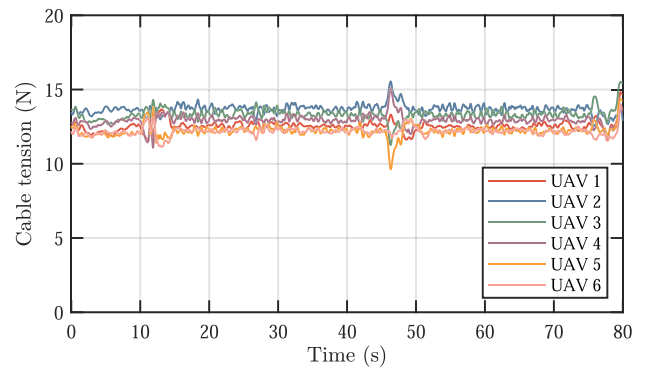


Fig. 10. Measured cable tensions during the outdoor experiment.

cable conditions. By constructing a passive system comprising interconnected virtual nodes, springs, and dampers, the framework achieves full decoupling between internal stability maintenance and external trajectory tracking. Numerical simulations involving ten UAVs demonstrate that the framework ensures balanced tension distribution and stable trajectory tracking even under extreme loading conditions—a 14 kg payload, representing over 76% of the collective thrust capacity. Field tests with six UAVs transporting a 6 kg load is conducted to validate the control framework's performance in practical applications. The

decentralized architecture, which relies solely on relative positioning and exhibits linear computational complexity, supports scalability for large-scale UAV clusters. Future research will explore more advanced challenges in networked systems, such as dynamic topology reconfiguration for obstacle avoidance, and security-guaranteed control under different disturbances.

APPENDIX A PROOF OF PROPOSITION 2

Substituting the control input (34) into the dynamic (32) gives:

$$m_i \dot{\mathbf{e}}_i = -\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} - (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i. \quad (51)$$

The Lyapunov function is constructed as:

$$V_S = T + V_E = \frac{1}{2} \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \mathbf{e}_i + \frac{1}{4} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \Delta \bar{l}_{ij}^2. \quad (52)$$

The omission of V_G in V_S is justified by the gravity compensation in the control input (34), which decouples gravitational effects from the error dynamics. Taking the derivative of the Lyapunov function (52) yields:

$$\dot{V}_S = \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \dot{\mathbf{e}}_i + \frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \Delta \bar{l}_{ij} \Delta \dot{\bar{l}}_{ij} \geq 0. \quad (53)$$

Substituting (51) into the term $\sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \dot{\mathbf{e}}_i$ results in:

$$\begin{aligned} \sum_{i=1}^{n+2} m_i \mathbf{e}_i^T \dot{\mathbf{e}}_i &= \sum_{i=1}^{n+2} \mathbf{e}_i^T [-\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)} - (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i] \\ &= \sum_{i=1}^{n+2} \mathbf{e}_i^T (-\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)}) \\ &\quad - \sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i. \end{aligned} \quad (54)$$

Since $(c_2 \mathbf{I} + \mathbf{K})$ is a positive definite matrix, we have:

$$-\sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i \leq 0. \quad (55)$$

Equality holds if and only if $\mathbf{e}_i = 0$, for $i = 1, 2, \dots, n+2$. Next, we analyze the term $\sum_{i=1}^{n+2} \mathbf{e}_i^T (-\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)})$ in (54).

$$\begin{aligned} &\sum_{i=1}^{n+2} \mathbf{e}_i^T (-\mathbf{F}_{s(\mathbf{q}_i)} - \mathbf{F}_{d(\mathbf{q}_i, \dot{\mathbf{q}}_i)}) \\ &= \sum_{i=1}^{n+2} (\mathbf{v}_i - \mathbf{v}_d)^T \left[-\sum_{j \in N_{n+2,i}} k_{ij} w_{ij} \left(1 - \frac{l_{ij0}}{l_{ij}}\right) (\mathbf{q}_i - \mathbf{q}_j) \right. \\ &\quad \left. - c_1 \sum_{j \in N_{n+2,i}} w_{ij} (\dot{\mathbf{q}}_i - \dot{\mathbf{q}}_j) \right] \\ &= -\frac{1}{2} \sum_{i,j=1}^{n+2} [(\mathbf{v}_i - \mathbf{v}_d) - (\mathbf{v}_j - \mathbf{v}_d)]^T \end{aligned}$$

$$\begin{aligned} &\times \left[k_{ij} w_{ij} \left(1 - \frac{l_{ij0}}{l_{ij}}\right) (\mathbf{q}_i - \mathbf{q}_j) + c_1 w_{ij} (\dot{\mathbf{q}}_i - \dot{\mathbf{q}}_j) \right] \\ &= -\frac{1}{2} \sum_{i,j=1}^{n+2} (\mathbf{v}_i - \mathbf{v}_j)^T \\ &\quad \times \left[k_{ij} w_{ij} \left(1 - \frac{l_{ij0}}{l_{ij}}\right) (\mathbf{q}_i - \mathbf{q}_j) + c_1 w_{ij} (\mathbf{v}_i - \mathbf{v}_j) \right] \\ &= -\frac{1}{2} \sum_{i,j=1}^{n+2} \left[k_{ij} w_{ij} \frac{l_{ij} - l_{ij0}}{l_{ij}} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{q}_i - \mathbf{q}_j) \right. \\ &\quad \left. + c_1 w_{ij} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{v}_i - \mathbf{v}_j) \right]. \end{aligned} \quad (56)$$

For the term $\frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \Delta \bar{l}_{ij} \Delta \dot{\bar{l}}_{ij}$ in (52), we have:

$$\begin{aligned} &\frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \Delta \bar{l}_{ij} \Delta \dot{\bar{l}}_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} (\bar{l}_{ij} - \bar{l}_{ij0}) \dot{\bar{l}}_{ij} \\ &= \frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} (\bar{l}_{ij} - \bar{l}_{ij0}) \frac{d}{dt} \|\mathbf{q}_i - \mathbf{q}_j\|_2 \\ &= \frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \frac{\bar{l}_{ij} - \bar{l}_{ij0}}{\bar{l}_{ij}} (\mathbf{q}_i - \mathbf{q}_j)^T (\dot{\mathbf{q}}_i - \dot{\mathbf{q}}_j) \\ &= \frac{1}{2} \sum_{i,j=1}^{n+2} k_{ij} w_{ij} \frac{\bar{l}_{ij} - \bar{l}_{ij0}}{\bar{l}_{ij}} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{q}_i - \mathbf{q}_j). \end{aligned} \quad (57)$$

By combining (53), (54), (55), and (57), it follows that

$$\begin{aligned} \dot{V}_S &= -\frac{1}{2} \sum_{i,j=1}^{n+2} c_1 w_{ij} (\mathbf{v}_i - \mathbf{v}_j)^T (\mathbf{v}_i - \mathbf{v}_j) \\ &\quad - \sum_{i=1}^{n+2} \mathbf{e}_i^T (c_2 \mathbf{I} + \mathbf{K}) \mathbf{e}_i \leq 0. \end{aligned} \quad (58)$$

Equality holds if and only if $\mathbf{e}_i = 0$, for $i = 1, 2, \dots, n+2$ and $\mathbf{v}_i - \mathbf{v}_j = 0$, for $i, j = 1, 2, \dots, n+2$. Since $V_S \geq 0$ and is radially unbounded, it follows from LaSalle's Invariance Principle that all system trajectories converge to the invariant set:

$$\begin{aligned} M_d &= \{\dot{\mathbf{q}} \mid \mathbf{e}_i = \mathbf{v}_i - \mathbf{v}_d = 0, \mathbf{v}_i - \mathbf{v}_j = 0\}, \\ i, j &= 1, 2, \dots, n+2. \end{aligned} \quad (59)$$

That is,

$$\begin{cases} \lim_{t \rightarrow \infty} \|\mathbf{e}_i(t)\| = \lim_{t \rightarrow \infty} \|\mathbf{v}_i(t) - \mathbf{v}_d(t)\| = 0 \\ \lim_{t \rightarrow \infty} \|\mathbf{v}_i(t) - \mathbf{v}_j(t)\| = 0, \quad i, j = 1, 2, \dots, n+2 \end{cases} \quad (60)$$

The physical interpretation is that the virtual passive system, as a whole, can asymptotically track the desired velocity $\mathbf{v}_d$, while the individual virtual nodes within the system remain relatively stationary with respect to one another. $\square$

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